

OR

Q.6	a.	What is deadlock? What are the necessary conditions for the deadlock to occur?	6	L1, L2	CO3																																																																																										
	b.	Consider the following snap-shot of a system: <table><tr><th rowspan="2">Process</th><th colspan="4">Allocation</th><th colspan="4">Max</th><th colspan="4">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>P0</td><td>2</td><td>0</td><td>0</td><td>1</td><td>4</td><td>2</td><td>1</td><td>2</td><td>3</td><td>3</td><td>2</td><td>1</td></tr><tr><td>P1</td><td>3</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P2</td><td>2</td><td>1</td><td>0</td><td>3</td><td>2</td><td>3</td><td>1</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td>1</td><td>3</td><td>1</td><td>2</td><td>1</td><td>4</td><td>2</td><td>4</td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td>1</td><td>4</td><td>3</td><td>2</td><td>3</td><td>6</td><td>6</td><td>5</td><td></td><td></td><td></td><td></td></tr></table> Answer the following using Banker's algorithm: i) Is the system in safe state? If so give the safe sequence. ii) If process P2 requests (0, 1, 1, 3) resource can it be granted immediately.	Process	Allocation				Max				Available				A	B	C	D	A	B	C	D	A	B	C	D	P0	2	0	0	1	4	2	1	2	3	3	2	1	P1	3	1	2	1	5	2	5	2					P2	2	1	0	3	2	3	1	6					P3	1	3	1	2	1	4	2	4					P4	1	4	3	2	3	6	6	5					14	L3	CO3
Process	Allocation				Max				Available																																																																																						
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P0	2	0	0	1	4	2	1	2	3	3	2	1																																																																																			
P1	3	1	2	1	5	2	5	2																																																																																							
P2	2	1	0	3	2	3	1	6																																																																																							
P3	1	3	1	2	1	4	2	4																																																																																							
P4	1	4	3	2	3	6	6	5																																																																																							

Module – 4

Q.7	a.	What is paging? Explain with neat diagram paging hardware with TLB?	10	L1, L2	CO4
	b.	What are the commonly used strategies to select a free hole from the available holes?	6	L1	CO4
	c.	Explain fragmentation in detail.	4	L2	CO4

OR

Q.8	a.	With a neat diagram? Describe the steps in handling the page fault.	8	L2	CO4
	b.	Consider the page reference string: 1, 0, 7, 1, 0, 2, 1, 2, 3, 0, 3, 2, 4, 0, 3, 6, 2, 1 for a memory with 3 frames. Determine the number of page faults using F1, F0, optimal and LRU replacement algorithms which algorithm is more efficient.	12	L3	CO4

Module – 5

Q.9	a.	Define file. List and explain the different file attributes and operations.	10	L1	CO5
	b.	Explain the different allocation methods.	10	L2	CO5

OR

Q.10	a.	What is Access Matrix? Explain Access Matrix method of system protection with domain as objects and its implementation.	10	L1, L2	CO5
	b.	A drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request 143 and previously serviced a request at 125. The queue of pending requests in FIFO order is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 starting from current head position. What is the total distance travelled (in cylinders) by disk arm to satisfy the requests using FCFS, SSTF, SCAN, LOOK and C-LOOK algorithm.	10	L3	CO5
